

Motor Datasheet

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	68.0	mm
Machine Inner Diameter	32.2	mm
Machine Thickness / Width	23.7	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	377.0	g
Rotor Mass	101.6	g
Stator Mass	275.4	g
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	689.9	g cm2

Electrical Properties

Parameter	Value	Unit
Supply Current ('RMS')	82.044	A
Phase Resistance	0.242	mOhm
D-Axis Inductance (Ld)	0.682	μH
Q-Axis Inductance (Lq)	0.648	μH
Terminal Inductance (Ltt)	1.330	μH
PM Flux Linkage	0.456	mWb
Frequency at operating point (electrical)	416.667	Hz
Winding Configuration	wye (star)	-

DC Supply Voltage	48.000	V
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Performance

Parameter	Value	Unit
Torque (Continuous)	800.0	mNm
Speed	2500.0	rpm
BackEMF Constant (Ke)	0.5	mV/rpm
Motor Constant (Km, only resistive losses)	362.1	mNm/sqrt(W)
Specific Motor Constant (Km / mass)	960.6	mNm/sqrt(W)/kg
Torque Constant (Kt, no-load)	6.8	mNm/A_peak
Torque Ripple	1.2	%
Saturation	0.0	%

Losses

Parameter	Value	Unit
Total Losses	5.9	W
Resistive Losses	4.9	W
Iron Core Losses	7.0	W
Magnet Eddy Current Losses	2.0	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-

Number of Slots	18	-
Pole Pairs	10	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	1.0	-
Number of Phase Turns (all phases in series)	6.0	-

Materials

Component	Material	
Stator Core	N030-Cogent	-
Magnets	BMN-52SH (GBD)	-